

Limits and Continuity

1. Evaluate the following limits, or state that they do not exist.

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x}, \quad \lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{x^2 + 5}, \quad \lim_{x \rightarrow 0^+} \frac{1}{x}, \quad \lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^2 - 4}.$$

2. Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Using only the intuitive idea that limits respect algebraic operations, justify each of the following:

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) + g(x)] &= L + M, & \lim_{x \rightarrow a} [f(x)g(x)] &= LM, \\ \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \frac{L}{M} \quad (M \neq 0), & \lim_{x \rightarrow a} c \cdot f(x) &= cL. \end{aligned}$$

Use these rules to evaluate $\lim_{x \rightarrow 2} (3x^2 - 5x + 1)$ and $\lim_{x \rightarrow 1} \frac{x^3 + 2x}{x^2 + 1}$.

3. Suppose $f(x) \leq g(x) \leq h(x)$ for all x near a , and suppose

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L.$$

- (a) Explain informally why $\lim_{x \rightarrow a} g(x) = L$ must hold. This is called the *Squeeze Theorem*.
- (b) Use the Squeeze Theorem to evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$. *Hint: what can you say about $|\sin(1/x)|$ for all $x \neq 0$?*
- (c) Use the Squeeze Theorem and the geometric inequality

$$\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1, \quad \theta \in (0, \frac{\pi}{2}),$$

to show that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$.

4. **Definition.** A function f is *continuous at a* if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Note this requires three things: $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and the two are equal.

For each function below, determine all points of discontinuity and classify each as *removable*, *jump*, or *infinite*.

$$f(x) = \frac{x^2 - 1}{x - 1}, \quad g(x) = \begin{cases} x + 1 & x < 0 \\ x^2 & x \geq 0 \end{cases}, \quad h(x) = \frac{1}{(x - 2)^2}, \quad k(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

5. Suppose f and g are continuous at a .
- (a) Show that $f + g$ and $f \cdot g$ are continuous at a .
 - (b) Is f/g necessarily continuous at a ? If not, give a counterexample and state a condition under which it is.
 - (c) Suppose additionally that g is continuous at $f(a)$. Show that the composition $g \circ f$ is continuous at a .

6. Let $f(x) = \frac{|x|}{x}$ for $x \neq 0$.

- (a) Compute $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$.
- (b) Explain why $\lim_{x \rightarrow 0} f(x)$ does not exist.
- (c) Show that f is continuous at every $a \neq 0$. *Hint: consider $a > 0$ and $a < 0$ separately.*
- (d) Can f be defined at $x = 0$ in a way that makes it continuous there? Explain.

7. (**Intermediate Value Theorem.**) Let f be continuous on a closed interval $[a, b]$, and let N be any value strictly between $f(a)$ and $f(b)$. Then there exists some $c \in (a, b)$ such that $f(c) = N$.

- (a) Use the Intermediate Value Theorem to show that $x^5 - 3x + 1 = 0$ has at least one root in $[1, 2]$ and at least one root in $[-2, -1]$.
- (b) Show that the equation $\cos x = x$ has at least one real solution. *Hint: consider $f(x) = \cos x - x$ on a suitable interval.*
- (c) Explain why every polynomial of odd degree with real coefficients must have at least one real root. *Hint: think about the behavior of odd-degree polynomials as $x \rightarrow \pm\infty$.*

8. Evaluate the following limits.

$$\lim_{x \rightarrow \infty} \frac{5x^3 - x}{2x^3 + 7x^2 - 1}, \quad \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x} - x \right), \quad \lim_{x \rightarrow -\infty} \frac{e^x}{1 + e^x}.$$

For the middle limit, multiply and divide by the conjugate $\sqrt{x^2 + x} + x$ and simplify before taking the limit.

9. Recall the difference quotient $A(f, h, x) = \frac{f(x+h) - f(x)}{h}$ from the Functions sheet.

- (a) Compute $A(f, h, 4)$ for $f(x) = \sqrt{x}$ and simplify fully. Then evaluate $\lim_{h \rightarrow 0} A(f, h, 4)$.
- (b) Repeat for $f(x) = \frac{1}{x}$ at general $x \neq 0$.
- (c) For which value(s) of x does the limit in part (b) fail to exist? What does this say about the graph of $f(x) = 1/x$?
- (d) Compute $\lim_{h \rightarrow 0} A(f, h, x)$ for $f(x) = x^n$, n a positive integer. *Hint: use the factorization $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$ with $a = x + h$, $b = x$.*

10. Suppose f is continuous on $[a, b]$.

- (a) Show that if $f(x) \neq 0$ for all $x \in [a, b]$, then f must be either entirely positive or entirely negative on $[a, b]$. *Hint: use the Intermediate Value Theorem.*
- (b) Suppose $f(a) < 0 < f(b)$. Show that the set of roots $\{x \in [a, b] : f(x) = 0\}$ is nonempty. Does it have to be finite? Give an example where it is infinite.
- (c) Give an example of a function on $[0, 1]$ that is *not* continuous but still satisfies: for every y_0 strictly between $f(0)$ and $f(1)$, there exists $c \in (0, 1)$ with $f(c) = y_0$. What does this show about the converse of the IVT?

11. Let $f(x) = \sin x$.

- (a) Using the identity

$$\sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right),$$

show that $|\sin x - \sin y| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

- (b) Conclude from part (a) that $\sin x$ is continuous at every $a \in \mathbb{R}$.
- (c) Using the analogous identity for cosine, show that $\cos x$ is also continuous everywhere.
- (d) Using parts (b) and (c) together with Problem 5, determine at which values of x the function $\tan x = \sin x / \cos x$ is continuous.